

GIS-Based Bus Depot Navigation System

Koyambedu CMBT, Chennai - Methodology

Methodology: The project was completed using a GIS-to-web mapping workflow. Important locations, the study-area boundary and navigation routes were prepared as GIS data and then displayed in an interactive Leaflet web map.

- 1. Study Area Selection:** Koyambedu CMBT, Chennai was selected as the bus-depot study area. Important facilities and transport-related locations within and around the terminus were identified.
- 2. Location Data Preparation:** Point locations were prepared with basic attributes such as ID, name, type, category, coordinates and description. The data was checked before use in the web map.
- 3. GIS Layer Creation:** ArcGIS Pro was used to prepare the point layer, project-area boundary and three navigation route lines.
- 4. GeoJSON Conversion:** The GIS layers were exported to GeoJSON format so that they could be read directly by the web application.
- 5. Web Map Development:** A website was created using HTML, CSS and JavaScript. LeafletJS was used for the interactive map and OpenStreetMap was used as the base map.
- 6. Map Display:** The GeoJSON point, boundary and route layers were loaded into the Leaflet map. Point popups, a scale bar and a location data table were added for easy viewing.
- 7. Testing and Publishing:** The website was tested locally using Live Server. After checking the map, popups, routes and table, the completed project was published using GitHub Pages for online sharing.

Tools Used

ArcGIS Pro	GIS data
HTML / CSS	Web page
JavaScript	Data loading
LeafletJS	Web map
GitHub Pages	Hosting

Final Output: A public web map showing the Koyambedu CMBT study area, important point locations, project boundary, navigation routes, point information and location data table.